

Context

BGP (Border Gateway Protocol) is the main routing protocol on the Internet, designed to interconnect networks operated by different ISPs, and is therefore one of the fundamental elements of the Internet. It is described as a path vector protocol, since BGP does not use traditional IGP metrics (OSPF, EIGRP, RIP), but makes routing decisions based on paths, network policies, and/or rules defined by operators. BGP maintains a table of IP networks or “prefixes” that designate the ability to access the network between autonomous systems (AS).

In this activity, we will configure external BGP (eBGP) and internal BGP (iBGP) routes using GNS3, as Packet Tracer only supports eBGP and does not allow all test scenarios to be implemented.

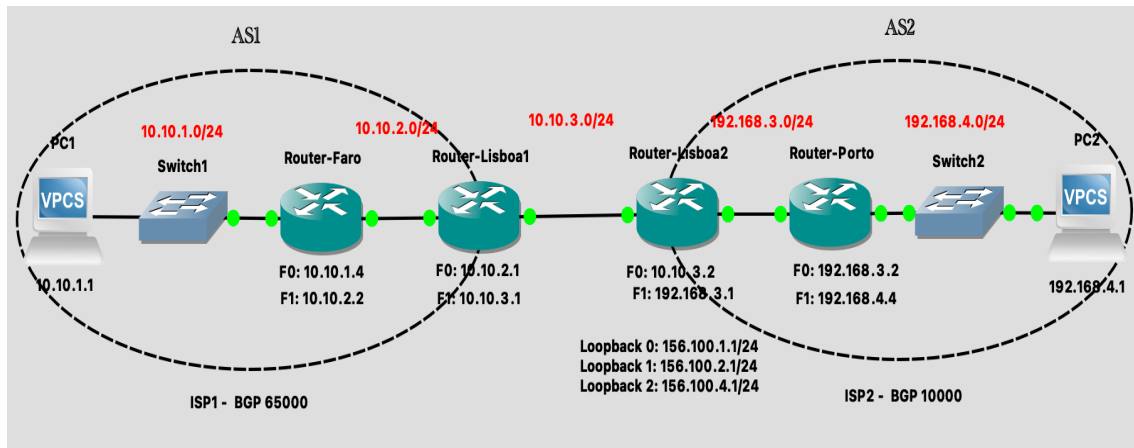
Useful links:

https://www.cisco.com/c/en/us/td/docs/ios/iproute_bgp/command/reference/irg_book/irg_bgp1.html

https://www.netwrix.com/cisco_commands_cheat_sheet.html

1 – Basic Topology

The base topology provided in the link is illustrated in the figure below. Some of the configurations have already been completed, but you should check that everything is correct. This topology uses the scheme already used in previous sessions, where two Autonomous Systems (ASs) belonging to two different ISPs (ISP1 and ISP2) are defined. Several loopback interfaces are configured on the ISP2 router to simulate connected networks that are advertised to the ISP1 router through BGP peering.



Start by checking that all IP addressing is properly configured and make sure that all interfaces are activated using the command “no shutdown”.

2 – Configuring Loopback Interfaces on ISP2

As mentioned in the description of the base topology, several interfaces are configured on the ISP2 border router, which simulate networks connected to this service provider that are to be advertised to ISP1. To do this, the three addresses indicated in the base topology will be configured with their respective VLSM masks.

```
Router-Lisboa2 (config)# interface Loopback0
Router-Lisboa2 (config-if)# ip address 156.100.1.1 255.255.255.0
Router-Lisboa2 (config)# interface Loopback1
Router-Lisboa2 (config-if)# ip address 156.100.2.1 255.255.255.0
Router-Lisboa2 (config)# interface Loopback2
Router-Lisboa2 (config-if)# ip address 156.100.4.1 255.255.254.0
```

3 - eBGP configuration on ISP1 and ISP2

We will start configuring the BGP protocol using the router bgp [ASN] command. Next, use the bgp log-neighbor-changes command to enable/disable logging messages that are generated when the status of a BGP neighbor changes. We also use the no synchronization command, indicating that in this configuration, iBGP routers do not need to synchronize iBGP with their internal routing protocol, such as EIGRP or OSPF. Synchronization can be disabled if all routers in the AS run BGP, or if there is no transit traffic from a different AS. Finally, the BGP neighbor relationship is defined with the neighbor remote-as command, which initiates the peering

connection and adds an entry to the BGP neighbor table with an autonomous system (AS) number that identifies the neighbor to the local autonomous system. The AS number of the configured neighbor is compared with the AS number received in the “Open Message,” and the peering session is only initiated if the numbers match.

eBGP- Router Lisboa1 (ISP1)

```
Router-Lisboa1 (config)# router bgp 65000
Router-Lisboa1 (config)# bgp log-neighbor-changes
Router-Lisboa1 (config)# no synchronization
Router-Lisboa1 (config)# neighbor 10.10.3.2 remote-as 10000
Router-Lisboa1(config-router)#network 10.10.2.0 mask 255.255.255.0
Router-Lisboa1(config-router)#network 10.10.3.0 mask 255.255.255.0
```

eBGP - Router Lisboa2 (ISP2)

```
Router-Lisboa2 (config)# router bgp 10000
Router-Lisboa2 (config-router)# bgp log-neighbor-changes
Router-Lisboa2 (config-router)# no synchronization
Router-Lisboa2 (config-router)# neighbor 10.10.3.1 remote-as 65000
Router-Lisboa2(config-router)#network 10.10.3.0 mask 255.255.255.0
Router-Lisboa2(config-router)#network 192.168.3.0 mask 255.255.255.0
Router-Lisboa2 (config-router)# network 156.100.4.0 mask 255.255.255.0
Router-Lisboa2 (config-router)# network 156.100.2.0 mask 255.255.255.0
Router-Lisboa2 (config-router)# network 156.100.1.0 mask 255.255.255.0
```

4 – Verification of eBGP Configuration

The confirmation of BGP external relations will be verified using three commands:

1. **show ip bgp:**
2. **show ip bgp summary**
3. **show ip neighbours**

```
Router-Lisboa1 # show ip bgp
```

```

Router-Lisboa1#sh ip bgp
BGP table version is 7, local router ID is 10.10.3.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop           Metric LocPrf Weight Path
*> 10.10.2.0/24    0.0.0.0             0         32768 i
* 10.10.3.0/24    10.10.3.2           0          0 10000 i
*>                 0.0.0.0             0         32768 i
*> 156.100.1.0/24  10.10.3.2           0          0 10000 i
*> 156.100.2.0/24  10.10.3.2           0          0 10000 i
*> 156.100.4.0/24  10.10.3.2           0          0 10000 i
*> 192.168.3.0     10.10.3.2           0          0 10000 i

```

Interpretation of the information shown:sh ip

Aug 14 18:36:29.287: %BGP-5-ADJCHANGE: neighbor 10.10.3.2 Up

This log message indicates that the BGP adjacency (neighborhood relationship) between “Router-Lisbon1” and the router with IP 10.10.3.2 (Lisbon2) has come up. This means that BGP has successfully established a connection with the neighboring router, and now the routers can exchange BGP routes.

Headers

BGP Table Version: The BGP table version is 4, indicating the number of times it has been updated.

Local Router ID: The router ID is 10.10.2.1. This is the unique identifier for the BGP process on “Router-Lisbon1.” It was assigned automatically because the router had an active interface with the IP 10.10.2.1 or was configured manually.

Status Codes: These codes describe the status of the routes:

* - The route is valid.

> - This route is the best route to reach the destination.

i - This route originated within the same autonomous system (AS) (internal BGP route).

Announced BGP Routes - column analysis:

- **Network:** This is the network destination that can be reached. For example, 156.100.1.0/24 refers to the network with a /24 mask.
- **Next Hop:** The IP address of the next router (next hop) that must be used to reach the destination network. In all cases here, the next hop is 10.10.3.2.
- **Metric:** In BGP, the metric is not generally used in decision making. It is shown as 0, which is common in BGP.
- **LocPrf (Local Preference):** This value is 0, which means that the route is not receiving any specific local preference. Local Preference is used within an AS to influence the preferred path for outgoing traffic.
- **Weight:** This is a Cisco-specific value used to influence outgoing traffic. Here, it is 0, which means that no specific preference is set.

- Path: This is the AS path. The path shows the sequence of autonomous systems that the route has traveled through. For example, 10000 i means that the route came from AS 10000, and the i indicates that the route originated within the same AS (internal BGP route).

```
Router-Lisboa1 # show ip bgp summary
```

```
Router-Lisboa1#sh ip bgp summary
BGP router identifier 10.10.3.1, local AS number 65000
BGP table version is 7, main routing table version 7
3 network entries using 351 bytes of memory
3 path entries using 156 bytes of memory
2/1 BGP path/bestpath attribute entries using 248 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 779 total bytes of memory
BGP activity 3/0 prefixes, 3/0 paths, scan interval 60 secs

Neighbor      V      AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
10.10.3.2      4      10000    9      8        7    0    0 00:05:34    5
```

This view allows you to supplement the previous information with the following data:

1) BGP Router Identification and Information

BGP Router Identifier: The BGP identifier for the router is 10.10.2.1. This is a unique identifier for the BGP process on this router. The router ID can be automatically assigned from the highest active IP on the interfaces or can be configured manually.

Local AS Number: The local Autonomous System (AS) number is 65000. This is the AS to which “Router-Lisboa1” belongs. BGP uses this AS number to distinguish between internal and external routing.

BGP Table Version: The BGP table version is 4, which means that the table has been updated four times since the router began participating in BGP route exchange.

2) Memory usage information

3 Network Entries: There are 3 network entries in the BGP table, using 351 bytes of memory.

3 Path Entries: There are 3 different paths stored, using 156 bytes of memory.

2/1 BGP Path/Bestpath Attribute Entries: This means that there are 2 path entries and 1 entry that is considered the best path, using 248 bytes of memory.

1 AS-PATH Entry: There is 1 entry in the AS-PATH, which records the sequence of ASes that the routes traversed to reach the destination, using 24 bytes of memory.

Total of 779 Bytes: The BGP process is using a total of 779 bytes of memory on “Router-Lisboa1”.

3) BGP Activity Statistics

BGP Activity: The router has learned 3 prefixes (or networks) from other routers and currently has 3 BGP paths stored. No prefixes or paths have been removed (as indicated by the zeros after the slash).

Scan Interval: The BGP router checks the routing table every 60 seconds. This is how often BGP updates the routes.

4) BGP Neighbor Information

Neighbor: The IP address of the BGP neighbor, in this case 10.10.3.2, refers to the router with which "Router-Lisbon1" is exchanging BGP routes.

V: The version of BGP in use. Here, both routers are using BGP version 4, currently the most common version.

AS: The AS number of the neighbor. Neighbor 10.10.3.2 belongs to AS 10000. This means that the router "Router-Lisbon1" is performing an external BGP route exchange (eBGP) because the ASes are different (65000 vs. 10000).

MsgRcvd and MsgSent: These values indicate the number of BGP messages received (26) and sent (25) between neighboring routers.

TblVer: The version of the BGP table known to the neighbor. Here, the value is 4, which indicates that this neighbor's route table has also been updated four times.

InQ and OutQ: These columns indicate the size of the input (InQ) and output (OutQ) queues for BGP messages. In this case, both queues are at 0, which means that there are no BGP messages pending processing between the routers.

Up/Down: This is the time during which the BGP connection has been active. In this example, the BGP relationship has been active for 21 minutes and 16 seconds (00:21:16).

State/PfxRcd: The status of the BGP neighbor is "Established," as indicated by the presence of a number of received prefixes (3). The router has received 3 route prefixes from neighbor 10.10.3.2, and the BGP connection is working correctly. If the code were "Active," it would mean that it is still trying to establish the BGP connection.

```
Router-Lisboa1 # show ip bgp neighbours
```

```
Router-Lisboa1#sh ip bgp neighbors
BGP neighbor is 10.10.3.2, remote AS 10000, external link
BGP version 4, remote router ID 156.100.4.1
BGP state = Established, up for 00:36:29
Last read 00:00:28, last write 00:00:28, hold time is 180, keepalive interval is 60 seconds
Neighbor capabilities:
  Route refresh: advertised and received(old & new)
  Address family IPv4 Unicast: advertised and received
Message statistics:
  InQ depth is 0
  OutQ depth is 0

      Sent      Rcvd
Opens:          1          1
Notifications:  0          0
Updates:         0          1
Keepalives:     39         39
Route Refresh:  0          0
Total:          40         41
Default minimum time between advertisement runs is 30 seconds

For address family: IPv4 Unicast
BGP table version 4, neighbor version 4/0
Output queue size : 0
Index 1, Offset 0, Mask 0x2
1 update-group member

      Sent      Rcvd
Prefix activity:
----
Prefixes Current:  0          3 (Consumes 156 bytes)
Prefixes Total:    0          3
Implicit Withdraw:  0          0
Explicit Withdraw:  0          0
Used as bestpath:  n/a        3
Used as multipath:  n/a        0

      Outbound   Inbound
Local Policy Denied Prefixes:
-----
Bestpath from this peer:    3      n/a
Total:                      3      0
Number of NLRI in the update sent: max 0, min 0

Connections established 1; dropped 0
Last reset never
Connection state is ESTAB, I/O status: 1, unread input bytes: 0
Connection is ECN Disabled, Minimum incoming TTL 0, Outgoing TTL 1
Local host: 10.10.3.1, Local port: 57414
Foreign host: 10.10.3.2, Foreign port: 179

Enqueued packets for retransmit: 0, input: 0  mis-ordered: 0 (0 bytes)
```

```
Event Timers (current time is 0x2AB8CC):
Timer      Starts    Wakeups    Next
Retrans     40         1          0x0
TimeWait    0           0          0x0
AckHold     38         36         0x0
SendWnd     0           0          0x0
KeepAlive   0           0          0x0
GiveUp      0           0          0x0
PmtuAger    0           0          0x0
DeadWait    0           0          0x0

iss: 3603692655  snduna: 3603693442  sndnxt: 3603693442  sndwnd: 15598
irs: 872775100  rcvnxt: 872775947  rcvwnd: 15538  delrcvwnd: 846

SRTT: 298 ms, RTT0: 314 ms, RTV: 16 ms, KRTT: 0 ms
minRTT: 4 ms, maxRTT: 300 ms, ACK hold: 200 ms
Flags: active open, nagle
IP Precedence value : 6

Datagrams (max data segment is 1460 bytes):
Rcvd: 46 (out of order: 0), with data: 39, total data bytes: 846
Sent: 77 (retransmit: 1, fastretransmit: 0, partialack: 0, Second Congestion: 0), with data: 39, total data bytes: 786
```

This command provides detailed information about the established BGP neighbor relations. Some of the information was already covered in the previous commands, so the focus will be only on the additional information:

BGP Neighbor Overview

- The BGP neighbor is 10.10.3.2, remote AS 10000, external link: Specifies the exact IP of the neighbor and the remote AS.
- Remote router ID 156.100.4.1: Identifies the BGP peer by its unique router ID.

Neighborhood Capabilities

- Routing Table Update Capability: Shows that route updates are supported by both routers (advertised and received).

Detailed Message Statistics

- Opens, Notifications, Updates, Keepalives: Disaggregates the exact number of each type of BGP message exchanged with the neighbor.

Prefix Activity

- Current Prefixes: Shows the number of prefixes received from this neighbor (3 prefixes).
- Used as bestpath: Confirms that the 3 prefixes are being used as the best path.

TCP Connection Details

- TCP Statistics: Adds detailed TCP metrics such as SRTT, RTTO, and the total number of bytes and datagrams sent/received.

Event Timers

- Event timers such as Retransmission, AckHold, etc.: New detailed information about the internal TCP timers used to manage the BGP session.

Datagram Statistics

- Datagrams sent/received: Adds specific counts of TCP segments sent and received, retransmissions, and any packet loss or congestion events.

5 – iBGP configuration on ISP1 and ISP2

At this stage, the goal is to configure internal BGP routes within AS1. The neighborhood definition is similar to that performed in eBGP, but the route-reflector-client command is used to broadcast to the internal network. This command implements a mechanism like that of the designated router in OSPF to avoid a “full mesh” of connections. Another method that can be used to define internal BGP routes is confederations.

iBGP between the Faro router and the Lisbon1 router:

```
Router-Faro (config)# router bgp 65000
Router-Faro (config)# bgp log-neighbor-changes
Router-Faro (config)# no synchronization
Router-Faro (config-router)# neighbor 10.10.2.1 remote-as 65000
Router-Faro (config-router)# network 10.10.1.0 mask 255.255.255.0
Router-Faro (config-router)# network 10.10.2.0 mask 255.255.255.0
```

Checking the settings on the Faro router

```
Router-Faro (config)# show running-config | section router bgp
```

```
[Router-Faro#show running-config | section router bgp
router bgp 65000
  no synchronization
  bgp log-neighbor-changes
  network 10.10.1.0 mask 255.255.255.0
  network 10.10.2.0 mask 255.255.255.0
  neighbor 10.10.2.1 remote-as 65000
  no auto-summary
```

iBGP between the Lisbon1 router and the Faro router:

```
Router-Lisboa1 (config)# router bgp 65000
Router-Faro (config)# bgp log-neighbor-changes
Router-Faro (config)# no synchronization
Router-Lisboa1 (config-router)# neighbor 10.10.2.2 remote-as 65000
Router-Lisboa1 (config-router)# network 10.10.2.0 mask 255.255.255.0
Router-Lisboa1 (config-router)# network 10.10.3.0 mask 255.255.255.0
```

Verification of settings on the Lisbon router1

```
Router-Lisboa1 (config)# show running-config | section router bgp
```

```
Router-Lisboa1(config-router)#do show running-config | section router bgp
router bgp 65000
  no synchronization
  bgp log-neighbor-changes
  network 10.10.2.0 mask 255.255.255.0
  network 10.10.3.0 mask 255.255.255.0
  neighbor 10.10.2.2 remote-as 65000
  neighbor 10.10.3.2 remote-as 10000
  no auto-summary
```

iBGP configuration on ISP2

iBGP between the Porto router and the Lisbon2 router:

```
Router-Porto (config)# router bgp 10000
Router- Porto (config)# bgp log-neighbor-changes
Router- Porto (config)# no synchronization
Router- Porto (config-router)# neighbor 192.168.3.1 remote-as 10000
Router- Porto (config-router)# network 192.168.3.0 mask 255.255.255.0
Router- Porto (config-router)# network 192.168.4.0 mask 255.255.255.0
```

```
Router-Lisboa2 (config)# router bgp 65000
Router-Lisboa2(config-router)# neighbor 192.168.3.2 remote-as 10000
Router- Porto (config-router)# network 10.10.3.0 mask 255.255.255.0
Router- Porto (config-router)# network 192.168.3.0 mask 255.255.255.0
```

Checking the settings on the Porto router

```
Router-Porto (config)# show running-config | section router bgp
```

```
[Router-Porto(config-router)#do show running-config | section router bgp
router bgp 10000
no synchronization
bgp log-neighbor-changes
network 192.168.3.0
network 192.168.4.0
neighbor 192.168.3.1 remote-as 10000
no auto-summary
```

Checking of settings on the Lisbon2 router

```
Router-Lisboa2 (config)# show running-config | section router bgp
```

```
Router-Lisboa2(config-router)#do show running-config | section router bgp
router bgp 10000
no synchronization
bgp log-neighbor-changes
network 10.10.3.0 mask 255.255.255.0
network 156.100.1.0 mask 255.255.255.0
network 156.100.2.0 mask 255.255.255.0
network 156.100.4.0 mask 255.255.255.0
network 192.168.3.0
neighbor 10.10.3.1 remote-as 65000
neighbor 192.168.3.2 remote-as 10000
no auto-summary
```

6 – Checking the routing tables

The routing table for the Lisbon1 router is shown below:

Router-Faro# show ip route

```
Router-Faro(config-router)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    156.100.0.0/24 is subnetted, 3 subnets
B       156.100.1.0 [200/0] via 10.10.3.2, 00:29:00
B       156.100.2.0 [200/0] via 10.10.3.2, 00:29:00
B       156.100.4.0 [200/0] via 10.10.3.2, 00:29:00
B       192.168.4.0/24 [200/0] via 10.10.3.2, 00:07:25
    10.0.0.0/24 is subnetted, 3 subnets
C       10.10.1.0 is directly connected, FastEthernet0/0
C       10.10.2.0 is directly connected, FastEthernet1/0
B       10.10.3.0 [200/0] via 10.10.2.1, 00:29:05
B       192.168.3.0/24 [200/0] via 10.10.3.2, 00:29:00
```

As can be seen, there are routes with the symbol “B” that indicate they have been learned by BGP.

Router-Lisboa1# show ip route

```
Router-Lisboa1(config-router)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    156.100.0.0/24 is subnetted, 3 subnets
B       156.100.1.0 [20/0] via 10.10.3.2, 00:43:48
B       156.100.2.0 [20/0] via 10.10.3.2, 00:43:48
B       156.100.4.0 [20/0] via 10.10.3.2, 00:43:48
B       192.168.4.0/24 [20/0] via 10.10.3.2, 00:09:57
    10.0.0.0/24 is subnetted, 3 subnets
B       10.10.1.0 [200/0] via 10.10.2.2, 00:31:36
C       10.10.2.0 is directly connected, FastEthernet0/0
C       10.10.3.0 is directly connected, FastEthernet1/0
B       192.168.3.0/24 [20/0] via 10.10.3.2, 00:42:27
```

Similarly, for the Lisboa2 router, we have:

```
Router-Lisboa2# show ip route
```

```
Router-Lisboa2(config-router)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    156.100.0.0/24 is subnetted, 3 subnets
C       156.100.1.0 is directly connected, Loopback0
C       156.100.2.0 is directly connected, Loopback1
C       156.100.4.0 is directly connected, Loopback2
B       192.168.4.0/24 [200/0] via 192.168.3.2, 00:10:22
    10.0.0.0/24 is subnetted, 3 subnets
B       10.10.1.0 [20/0] via 10.10.3.1, 00:32:00
B       10.10.2.0 [20/0] via 10.10.3.1, 00:44:13
C       10.10.3.0 is directly connected, FastEthernet0/0
C       192.168.3.0/24 is directly connected, FastEthernet1/0
Router-Lisboa2(config-router)#
```

And finally, for the Porto router

```
Router-Porto(config-router)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    156.100.0.0/24 is subnetted, 3 subnets
B       156.100.1.0 [200/0] via 192.168.3.1, 00:10:47
B       156.100.2.0 [200/0] via 192.168.3.1, 00:10:47
B       156.100.4.0 [200/0] via 192.168.3.1, 00:10:47
C       192.168.4.0/24 is directly connected, FastEthernet1/0
    10.0.0.0/24 is subnetted, 3 subnets
B       10.10.1.0 [200/0] via 10.10.3.1, 00:10:42
B       10.10.2.0 [200/0] via 10.10.3.1, 00:10:42
B       10.10.3.0 [200/0] via 192.168.3.1, 00:10:47
C       192.168.3.0/24 is directly connected, FastEthernet0/0
Router-Porto(config-router)#
```

7 – Viewing BGP Neighbor Relationships

Using the following commands, you can obtain a complete diagnosis of the configuration performed:

- show ip bgp
- show ip bgp summary
- show ip bgp neighbors

Router Faro

```
Router-Faro(config-router)#do sh ip bgp
BGP table version is 12, local router ID is 10.10.2.2
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop          Metric LocPrf Weight Path
*> 10.10.1.0/24      0.0.0.0              0         32768 i
* i10.10.2.0/24     10.10.2.1             0        100      0 i
*>                  0.0.0.0              0         32768 i
*>i10.10.3.0/24     10.10.2.1             0        100      0 i
*>i156.100.1.0/24   10.10.3.2             0        100      0 10000 i
*>i156.100.2.0/24   10.10.3.2             0        100      0 10000 i
*>i156.100.4.0/24   10.10.3.2             0        100      0 10000 i
*>i192.168.3.0      10.10.3.2             0        100      0 10000 i
*>i192.168.4.0      10.10.3.2             0        100      0 10000 i
```

```
Router-Faro(config-router)#do sh ip bgp summ
BGP router identifier 10.10.2.2, local AS number 65000
BGP table version is 12, main routing table version 12
8 network entries using 936 bytes of memory
9 path entries using 468 bytes of memory
4/3 BGP path/bestpath attribute entries using 496 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1924 total bytes of memory
BGP activity 9/1 prefixes, 16/7 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
10.10.2.1      4 65000    107    103     12    0   0 01:11:24        7
```

Router Lisboa1

```
Router-Lisboa1(config-router)#do show bgp
BGP table version is 11, local router ID is 10.10.2.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete
```

Network	Next Hop	Metric	LocPrf	Weight	Path
*>i10.10.1.0/24	10.10.2.2	0	100	0	i
* i10.10.2.0/24	10.10.2.2	0	100	0	i
*>	0.0.0.0	0		32768	i
* 10.10.3.0/24	10.10.3.2	0		0	10000 i
*>	0.0.0.0	0		32768	i
*> 156.100.1.0/24	10.10.3.2	0		0	10000 i
*> 156.100.2.0/24	10.10.3.2	0		0	10000 i
*> 156.100.4.0/24	10.10.3.2	0		0	10000 i
*> 192.168.3.0	10.10.3.2	0		0	10000 i
*> 192.168.4.0	10.10.3.2	0		0	10000 i

```
Router-Lisboa1(config-router)#do show bgp summary
BGP router identifier 10.10.2.1, local AS number 65000
BGP table version is 11, main routing table version 11
8 network entries using 936 bytes of memory
10 path entries using 520 bytes of memory
5/4 BGP path/bestpath attribute entries using 620 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2100 total bytes of memory
BGP activity 8/0 prefixes, 12/2 paths, scan interval 60 secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
10.10.1.1	4	65000	0	0	0	0	0	never	Active
10.10.2.2	4	65000	104	108	11	0	0	01:12:40	2
10.10.3.2	4	10000	268	269	11	0	0	04:20:39	6

Router Lisboa2

```
Router-Lisboa2(config-router)#do show ip bgp
BGP table version is 13, local router ID is 156.100.4.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete
```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.10.1.0/24	10.10.3.1			0	65000 i
*> 10.10.2.0/24	10.10.3.1	0		0	65000 i
*> 10.10.3.0/24	0.0.0.0	0		32768	i
*> 10.10.3.0/24	10.10.3.1	0		0	65000 i
*> 156.100.1.0/24	0.0.0.0	0		32768	i
*> 156.100.2.0/24	0.0.0.0	0		32768	i
*> 156.100.4.0/24	0.0.0.0	0		32768	i
* i192.168.3.0	192.168.3.2	0	100	0	i
*> 192.168.3.0	0.0.0.0	0		32768	i
*> i192.168.4.0	192.168.3.2	0	100	0	i

```
Router-Lisboa2(config-router)#do show ip bgp sum
Router-Lisboa2(config-router)#do show ip bgp sum
BGP router identifier 156.100.4.1, local AS number 10000
BGP table version is 13, main routing table version 13
8 network entries using 936 bytes of memory
10 path entries using 520 bytes of memory
5/4 BGP path/bestpath attribute entries using 620 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2100 total bytes of memory
BGP activity 8/0 prefixes, 11/1 paths, scan interval 60 secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
10.10.3.1	4	65000	271	270	13	0	0	04:22:52	3
192.168.3.2	4	10000	97	103	13	0	0	01:31:03	2
192.168.4.1	4	10000	0	0	0	0	0	never	Active

Router do Porto

```
Router-Porto(config-router)#do sh ip bgp
BGP table version is 10, local router ID is 192.168.4.4
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete
```

Network	Next Hop	Metric	LocPrf	Weight	Path
*>i10.10.1.0/24	10.10.3.1	0	100	0	65000 i
*>i10.10.2.0/24	10.10.3.1	0	100	0	65000 i
*>i10.10.3.0/24	192.168.3.1	0	100	0	i
*>i156.100.1.0/24	192.168.3.1	0	100	0	i
*>i156.100.2.0/24	192.168.3.1	0	100	0	i
*>i156.100.4.0/24	192.168.3.1	0	100	0	i
* i192.168.3.0	192.168.3.1	0	100	0	i
*>	0.0.0.0	0		32768	i
*> 192.168.4.0	0.0.0.0	0		32768	i

```
Router-Porto(config-router)#do sh ip bgp su
BGP router identifier 192.168.4.4, local AS number 10000
BGP table version is 10, main routing table version 10
8 network entries using 936 bytes of memory
9 path entries using 468 bytes of memory
4/3 BGP path/bestpath attribute entries using 496 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1924 total bytes of memory
BGP activity 9/1 prefixes, 10/1 paths, scan interval 60 secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
192.168.3.1	4	10000	104	98	10	0	0	01:32:07	7


```

Router-Faro(config-router)#do sh ip bgp summ
BGP router identifier 10.10.2.2, local AS number 65000
BGP table version is 12, main routing table version 12
8 network entries using 936 bytes of memory
9 path entries using 468 bytes of memory
4/3 BGP path/bestpath attribute entries using 496 bytes of memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1924 total bytes of memory
BGP activity 9/1 prefixes, 16/7 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down  State/PfxRcd
10.10.2.1      4 65000    107    103     12   0   0 01:11:24      7

```

8 – Checking connectivity between PCs

It shows that end-to-end connectivity is ensured. It should be noted that in this exercise, BGP was used exclusively for internal and external routing, so one might question whether OSPF is necessary. The answer is that, although it is theoretically possible to use only BGP, it is not optimized for internal routing, resulting in slower convergence and greater consumption of memory and CPU resources.

```

PC1> ping 192.168.4.1

84 bytes from 192.168.4.1 icmp_seq=1 ttl=60 time=70.721 ms
84 bytes from 192.168.4.1 icmp_seq=2 ttl=60 time=63.197 ms
84 bytes from 192.168.4.1 icmp_seq=3 ttl=60 time=78.420 ms
84 bytes from 192.168.4.1 icmp_seq=4 ttl=60 time=63.112 ms
84 bytes from 192.168.4.1 icmp_seq=5 ttl=60 time=73.823 ms

```